

fix-design-notes

BOB-166 — fix design notes (scoping only, nothing implemented)

Captured 2026-08-21 while the defect was fresh. This is a SCOPE for whoever takes the item, not a patch. Everything below is read from source; nothing was changed.

Where the hole is

constitution/scripts/workable-items/cmd/workable-items/mutate.go

```
main.go:123      case "update":  runUpdate(args[1:])
mutate.go:69      func runUpdate(...)
mutate.go:152     if set["status"] {
mutate.go:153         ns := normalizeStatus(*status)
mutate.go:154         // normalizeStatus never returns a non-
closed-set value, but guard
mutate.go:155         // explicitly so a typo'd --status surfaces
rather than silently
mutate.go:156         // mapping to Queued. ...
mutate.go:158         cur.Status = ns
mutate.go:159     }
```

The validation asks exactly one question — *is this a legal §11.4.15 value?* — and a terminal status IS legal. Nothing asks the second question: *is this status legal FOR THIS ROW'S LOCATION?* So Fixed (→ Fixed.md) is accepted on a row whose `current_location` is still Issues, producing the state §11.4.19 forbids.

The sibling that already gets this right

move documents precisely the invariant update lacks
(mutate.go:18-21):

- move – the general REVERSE-OF-CLOSE relocation ... with an optional new status ... Refuses any destination/status pair that would create an INTEG-03 desync.

So the codebase already has the CONCEPT (a destination/status pair can be invalid) and a name for the failure (INTEG-03 desync). update simply never consults it. That matters for the fix: this is reusing an existing invariant at a seam that skipped it, NOT inventing a new rule (§11.4.227 — an anchor's done state is its SEAM landing, not its text landing).

Shape of the fix

1. In `runUpdate`, after `ns := normalizeStatus(*status)`, refuse when `ns` is terminal AND the resulting location would be `Issues`. The message must name the correct path rather than just saying no:

```
update: refusing to set terminal status "Fixed (→ Fixed.md)" on
BOB-NNN
while it is located in Issues – a terminal status without the
migration
leaves the row in the OPEN tracker (§11.4.19 requires the move
to be
atomic). Use instead:
  workable-items close BOB-NNN --db <p> --status fixed --
evidence <path>
or, if the row is genuinely already closed elsewhere:
  workable-items move --id BOB-NNN --db <p> --to Fixed --why
<text>
```

Naming both paths matters: `close` is right for a NEW closure (it demands evidence), `move` for reconciling an already-closed row.

2. Extend `validate` with a `status↔location` coherence invariant so the EXISTING ten rows are caught, not just future writes. A guard on the write seam alone leaves the current corpus silently wrong — the §11.4.146(D3) full-table-sweep argument: a diff-only check misses rows written before the guard existed, and raw-SQL writers bypass the CLI entirely.

Test-first obligations (§11.4.224)

- RED before the guard: a test that performs the exact reproduction (`update --status 'Fixed (→ Fixed.md)'` on a Queued row in `Issues`) and asserts the resulting row is NOT in the forbidden state. It must FAIL on today's code — verify that, do not assume it.

- Paired §1.1 mutation: remove the refusal; the test must go red. A guard never observed failing is unvalidated instrumentation (§11.4.115(F)).
- NEGATIVE CONTROL, non-negotiable (§11.4.201(1)): update -- status on a NON-terminal value must still work, and a terminal status on a row ALREADY in Fixed must still work — otherwise the guard becomes a false-positive refusal that blocks legitimate edits, which is a FAIL-bluff of equal severity to the hole it closes.

Constraint that is easy to miss

This code lives in the CONSTITUTION submodule and is consumed by other projects (§11.4.28). The fix must carry NO boba literal — no BOB- prefix, no docs/Issues.md path. It is a statement about the engine's own closed-set semantics, which is what makes it safe to land there. Per §11.4.26 the submodule must be fetched and pulled BEFORE editing, and pushed to all its upstreams after.

Evidence already captured

- evidence_absence_measurement_20260821.log — all ten rows have ZERO evidence rows, control-needed against two rows closed via close (1 each). Splits the drain into two classes: BOB-087/BOB-129 have unrecorded artifacts on disk; eight have nothing.